SENYAS: An Offline Intelligent System for Trilingual Sign Language and Braille Translation Using a Kinematic Hand Model, Landmark Geometry, and Braille Grid Fitting

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INTRODUCTION

Background of the Study

- RA 11106 (2018) made Filipino Sign Language the national sign language of the Filipino Deaf, but interpreters are rarely present.
- Most translation apps work in English, few in Filipino, almost none in Bikol, and they need the internet.
- Disasters in Albay (Mayon, typhoons, earthquakes) cut mobile data exactly when people must communicate.
- Deep sign recognisers need large video datasets that FSL does not have; braille readers assume scanners.

Table 1. The problem in figures

Indicator	Value	Source
People needing hearing care	> 430 million	WHO (2021)
People with vision impairment	≥ 2.2 billion	WHO (2019)
National sign language	FSL	RA 11106 (2018)
Languages of Albay homes	Bikol, Filipino, English	—
Offline FSL + braille app in Bikol	none found	Review, Ch. 2

Purpose of the study. To build and test SENYAS, an offline intelligent system that translates between sign language, braille, text and speech in English, Filipino and Bikol.

RRL/RRS

Table 2. Related literature and how the study uses it

Area	Key sources	Use in the present study
Sign language structure	Stokoe (1960); RA 11106 (2018)	Handshape, location and movement guide the recogniser
Hand tracking	Lugaresi et al. (2019); Zhang et al. (2020)	21 landmarks per hand, on the device
Robot kinematics	Craig (2005); Aristidou & Lasenby (2011)	Hand model and FABRIK thumb; avatar arms
Pattern recognition	Cover & Hart (1967); Sakoe & Chiba (1978); Schmitt (1938)	Nearest neighbour, DTW, hysteresis
Braille and images	ICEB (2013); Crow (1984); Viola & Jones (2001)	Grade 1 decoding; fast local contrast
Accessible software	W3C (2018); ISO/IEC 25010 (2011); CAST (2018)	Contrast, reflow, quality model, UDL lessons

Synthesis. Phones can track hands offline, and kinematics and geometry can describe hands and braille cells, so examples can be generated instead of collected.

RRL/RRS

Table 3. Related studies and their relation to the present study

Study	What was done	Relation to the present study
Cabalfin et al. (2012)	FSL recognition from video (manifold learning)	FSL can be recognised; SENYAS adds offline, Bikol, braille
Koller (2020); Rastgoo et al. (2021)	Surveys: deep, data-hungry methods	Motivates a geometric method without big data
Mitra & Acharya (2007)	Survey of gesture recognition	Separate handshape from movement
Antonacopoulos & Bridson (2004)	Braille from scanned pages	SENYAS reads phone photos and live video
Isayed & Tahboub (2015)	Review of optical braille recognition	Grid fitting and skew are the key steps

Synthesis. No free tool joins sign, braille, text and speech offline in English, Filipino and Bikol; the gap is an explainable, offline intelligent system.

THEORETICAL/CONCEPTUAL FRAMEWORK PARADIGM

Theoretical Framework

Research gap	Theory or method that addresses it
G1 No FSL dataset	Robot-hand kinematics generates labelled handshapes (FK, FABRIK).
G2 Hands vary in place, size, angle	Palm-frame coordinates are invariant to translation, scale, rotation.
G3 Signs move at different speeds	Hysteresis counting, winding number, dynamic time warping.
G4 Braille photos are skewed and noisy	Summed-area contrast, robust pitch and skew, grid fitting.

Figure 1. Research gaps (left) and the theories and methods used to address them (right)

THEORETICAL/CONCEPTUAL FRAMEWORK PARADIGM

Conceptual Paradigm

INPUT

- Camera frames of hands
- Braille photos or frames
- Typed or spoken text (EN, FIL, BIK)
- Kinematic hand model
- Trilingual lexicon (93 words, 32 phrases)

PROCESS

1. MediaPipe 21-point landmarks
2. Palm-frame features + nearest template
3. Motion rules, two-hand rules, DTW
4. Braille dots → grid fit → Grade 1 decode
5. Longest-match phrase lookup
6. Nine computational experiments

OUTPUT

- SENYAS app, website, Android app
- Text + Read aloud (on/off)
- Signing avatar + braille line
- Accuracy, CER, speed, layout results

Feedback: results guide a field validation with Deaf and blind learners

Figure 2. Conceptual paradigm of the study

SIGNIFICANCE OF THE STUDY

Deaf and hard-of-hearing learners. Their signing becomes spoken Filipino, English or Bikol, free and offline.

Blind and low-vision persons. Braille labels and notices are read aloud from a photo.

Teachers, SPED centres, families. Lessons, quizzes and camera practice in the home language.

Responders and health workers. An emergency phrasebook that works with no signal.

LGUs and DepEd. A free tool that supports RA 11106 and RA 7277.

Future researchers. An explainable method and open, repeatable benchmarks.

The study supports SDG 4 (inclusive education), SDG 10 (reduced inequalities) and SDG 11.5 (disaster resilience).

OBJECTIVES OF THE STUDY / SOP

General objective. To design, develop and evaluate SENYAS, an offline intelligent system for trilingual sign language and braille translation with a learning module.

Specifically, it seeks to answer the following questions:

1. How accurately are the 24 static letters classified as pose noise rises, and which design parts matter?
2. How reliably are 13 moving and 6 two-handed signs detected, with what false-alarm rate?
3. How accurately are fingerspelled words read across hold times and signer speeds?
4. How reliably are user-taught signs recognised, and how often are others falsely accepted?
5. How accurately are braille photos read across dot type, skew, blur and size?
6. How long does each component take per frame?
7. Does the interface fit all screens with readable contrast, and does the lexicon cover the phrasebook?

HYPOTHESES

Null Hypotheses

The hypotheses were stated before testing and are tested at the 0.05 level of significance.

Code	Hypothesis	Test or decision rule
H_o1	Static-letter accuracy at expected noise is at most 90%.	One-sided exact binomial test
H_o2	Detection of moving and two-handed signs is at most 90%.	One-sided exact binomial test
H_o3	Fingerspelled-word accuracy at the default hold time is at most 80%.	One-sided exact binomial test
H_o4	The braille character error rate is at least 20%.	Bound comparison; Wilson CI
H_o5	Processing time per frame exceeds 100 ms.	Median latency
H_o6	The interface has layout faults on at least one screen size.	Automated layout test

Reject H_o when $p < 0.05$ (or when the measured value is beyond the bound).

SCOPE AND DELIMITATION

Scope

- Alphabet A–Z with Ñ, numbers 0–10, 21 live signs
- 93 words, 32 phrases in EN, FIL, BIK
- User-taught signs; Grade 1 braille with Ñ
- Web, installable and Android apps

Delimitation

- No full FSL sentences or facial grammar
- No Grade 2 braille
- No tests with human participants in this phase
- Results from generated data (repeatable)

Table 4. Validation experiments of SENYAS

Experiment	Data	Measure
E1–E2 Static letters	24 letters × 40 hands × 4 noise levels; ablation	Accuracy
E3–E5 Moving, words, taught	19 signs × 30; 40 words × 9 settings; 4 taught signs	Detection, word accuracy
E6–E9 Braille, camera, speed, screens	300 photos; 288 renders; 11 screen sizes	CER, ms, layout faults

METHODOLOGY

Research Design

A developmental engineering-design study (Hevner et al., 2004). SENYAS is designed from kinematic and geometric models, built as working software, and evaluated with nine repeatable computational experiments.

1. Model → **2. Build** → **3. Benchmark** → **4. Analyse**

Table 5. Variables of the study

Independent (varied)	Dependent (measured)	Controlled (fixed)
Pose noise level; sign; hold time and signer speed	Accuracy; detection and false-alarm rates	Hand model and templates; random seed
Braille dot type, skew, blur, size	Word accuracy; character error rate	Frame rate 30 fps; image size
Screen size	Time per frame; layout faults; contrast	Same code as the shipped app

METHODOLOGY

Materials and Device

System specification

- Input: phone or laptop camera, microphone, keyboard, photos
- Engine: MediaPipe Hand Landmarker (on device)
- Recogniser: 572 templates × 41 features
- Braille reader: grid fitting, Grade 1 + Ñ
- Output: text in EN/FIL/BIK, Read aloud, avatar, braille
- App: 190 KB page + 17 MB offline engine; Android app



Table 6. Materials and components

Item	Specification	Purpose
Computer	x86-64, Linux, CPU only	Build and test
Chromium + Playwright	Headless browser	App, MediaPipe, braille, layout tests
Node.js 22	JavaScript runtime	Recogniser tests
MediaPipe 0.10.14	Float16 hand model	Landmarks
HTML/CSS/JS	No framework	App and website
Kotlin WebView	Android 7+	Android app
Python 3	Matplotlib, docx	Analysis, reports

METHODOLOGY

Data Gathering Procedure

Step 1

Define 37 handshapes with the hand model

Step 2

Generate 572 templates

Step 3

Build app, website, Android app

Step 4

E1 static letters (3,840 hands)

Step 5

E2 ablation (5 designs)

Step 6

E3 moving and two-handed signs

Step 7

E4 fingerspelled words

Step 8

E5 taught signs

Step 9

E6 braille photos (300)

Step 10

E7–E9 camera, speed, screens

Figure 3. Data gathering procedure (fixed random seeds, so every run can be repeated)

METHODOLOGY

Treatment of Data

Formulas

Accuracy = correct ÷ trials

Wilson CI: $(\hat{p} + z^2/2n \pm z\sqrt{(\hat{p}(1-\hat{p})/n + z^2/4n^2)}) \div (1 + z^2/n)$

Binomial test: $p = P(X \geq k \mid n, p_o)$

CER = Σ Levenshtein(read, true) ÷ Σ length(true)

Palm frame: $Y = (p_9 - p_o)/s$, $Z = (p_5 - p_{17}) \times Y$, $X = Y \times Z$

Grid fit: min over P, x_o of $\Sigma \min(|u - kP|, |u - \delta - k'P|)$

Statistical treatment

SOP	Question	Tool
1	Static-letter accuracy; ablation	Accuracy, Wilson CI, binomial
2	Moving and two-handed signs	Detection rate, binomial
3	Fingerspelled words	Word accuracy, CER, binomial
4	Taught signs	Detection, false accepts
5	Braille photos	Exact rate, CER
6	Speed	Median ms
7	Screens, contrast, coverage	Faults, WCAG ratio

α = 0.05; software: Node.js, Chromium, Python

METHODOLOGY

Ethical Considerations and Safety

Ethics. No human participants in this phase; all data were generated.

Privacy. Video, photos and audio never leave the device; consent before storage (RA 10173).

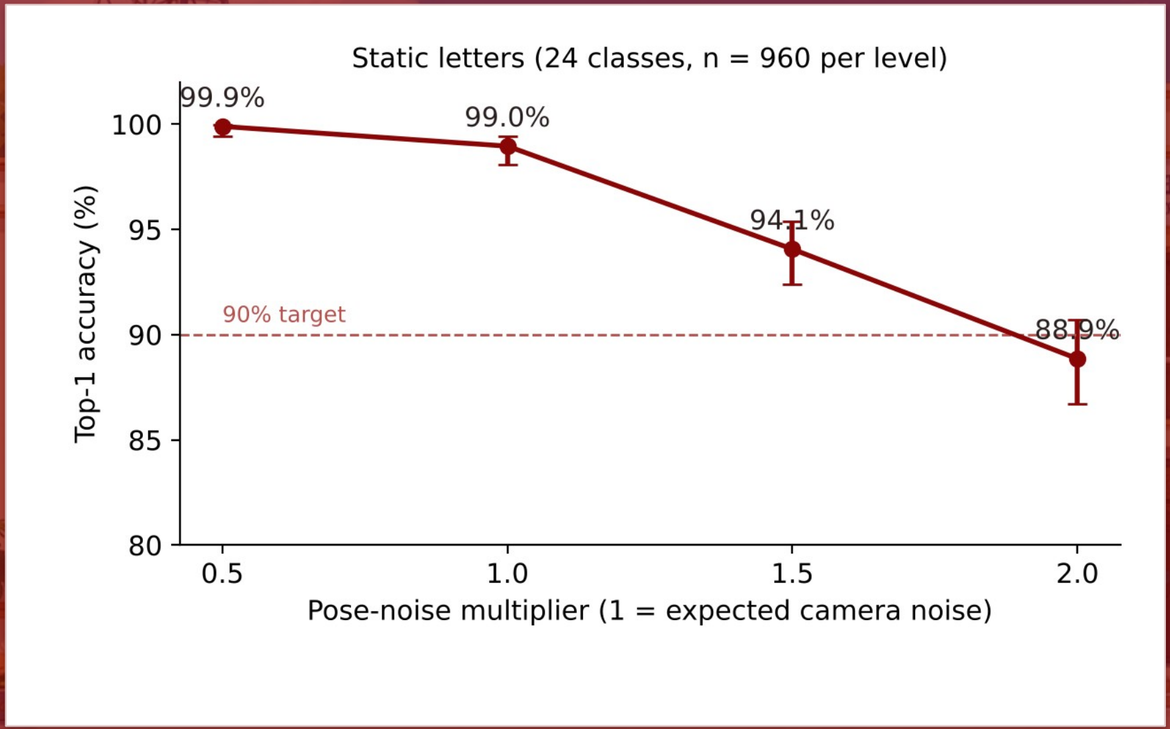
Honesty. All sources cited in APA 7; every number comes from open scripts.

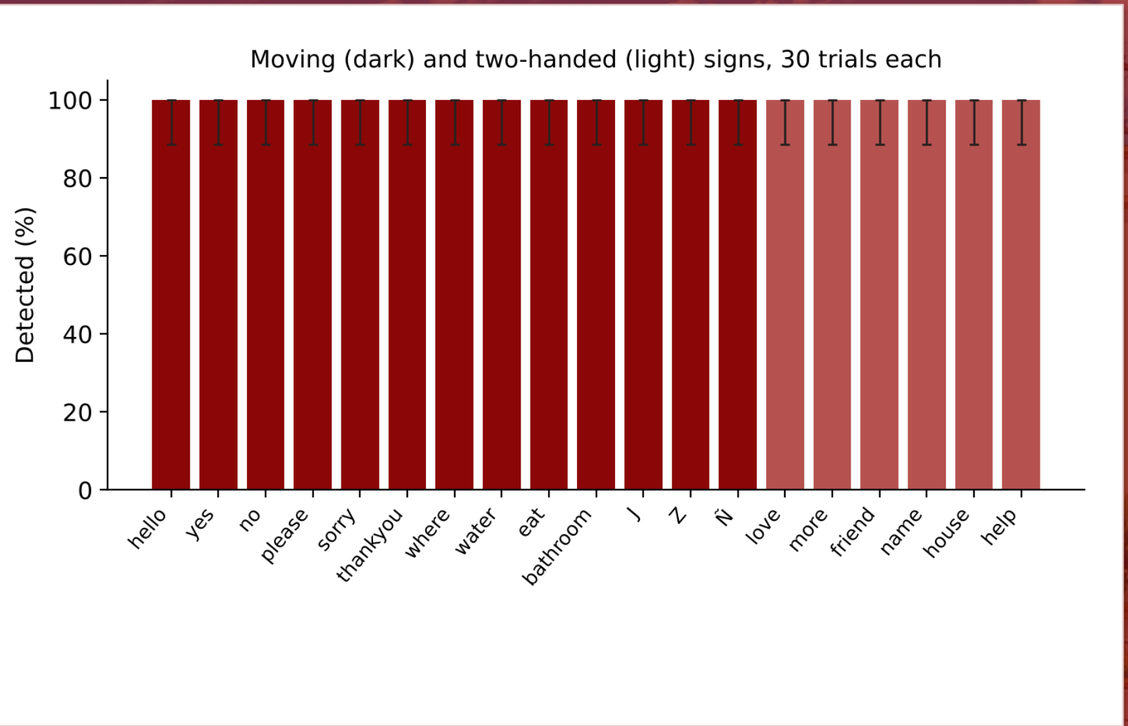
Accuracy notice. Signs shown are forms shared by FSL and ASL; local signs vary; not a certified interpreter.

Future tests. Ethics review, parental consent for minors, partnership with the Deaf community.

Licences. MediaPipe used under Apache-2.0.

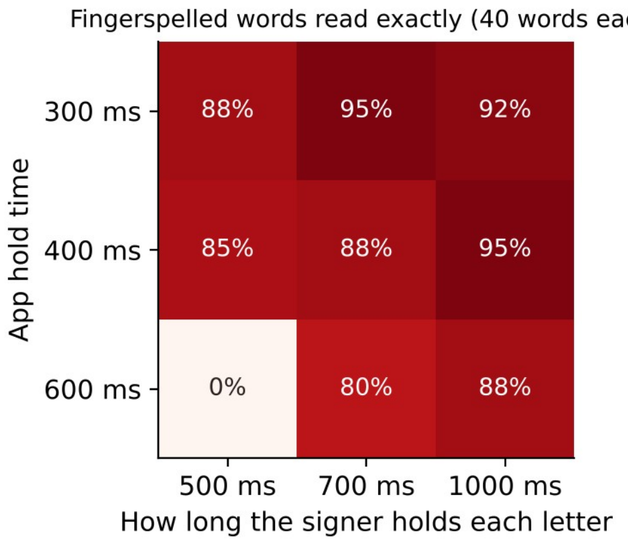
RESULTS AND DISCUSSION

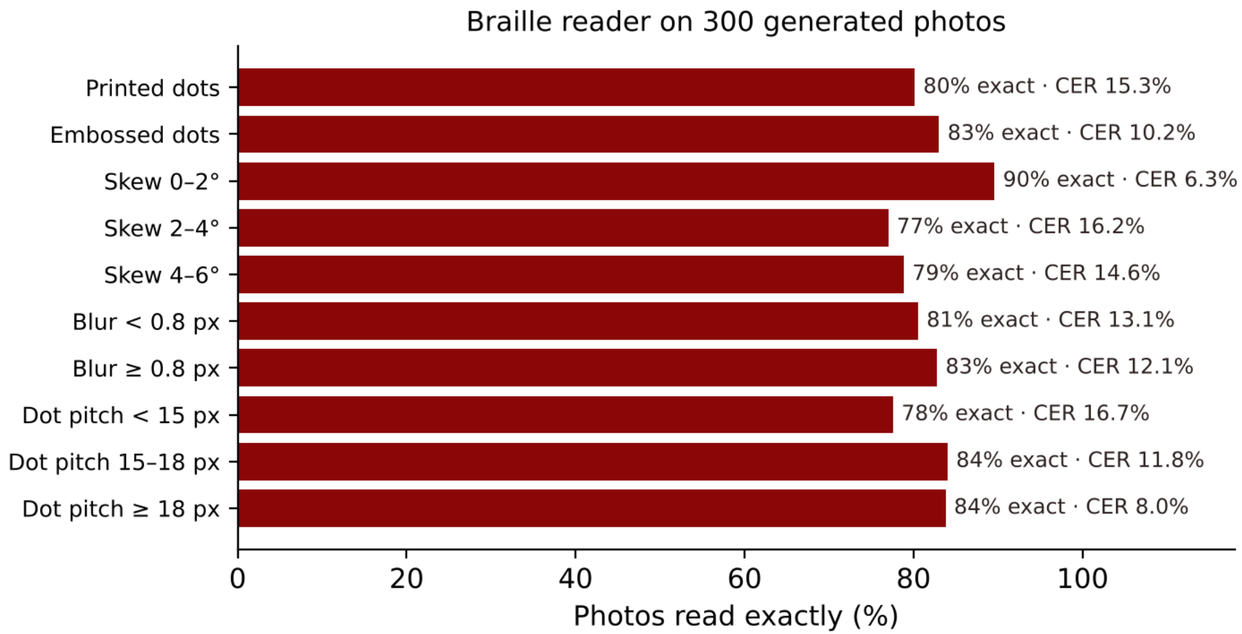




Result	Value	95% CI	Decision
Static letters at expected noise (E1)	99.0%	98.1%–99.4%	H_o1 rejected
Moving and two-handed signs (E3)	570/570	99.3%–100.0%	H_o2 rejected; 0 false alarms

RESULTS AND DISCUSSION





Result	Value	Decision
Fingerspelled words at 400 ms (E4)	89.2%	H_o3 rejected
Taught signs (E5)	120/120, 0 false accepts	Reliable from 3 takes
Braille photos (E6)	81.7% exact, CER 12.6%	H_o4 rejected

RESULTS AND DISCUSSION

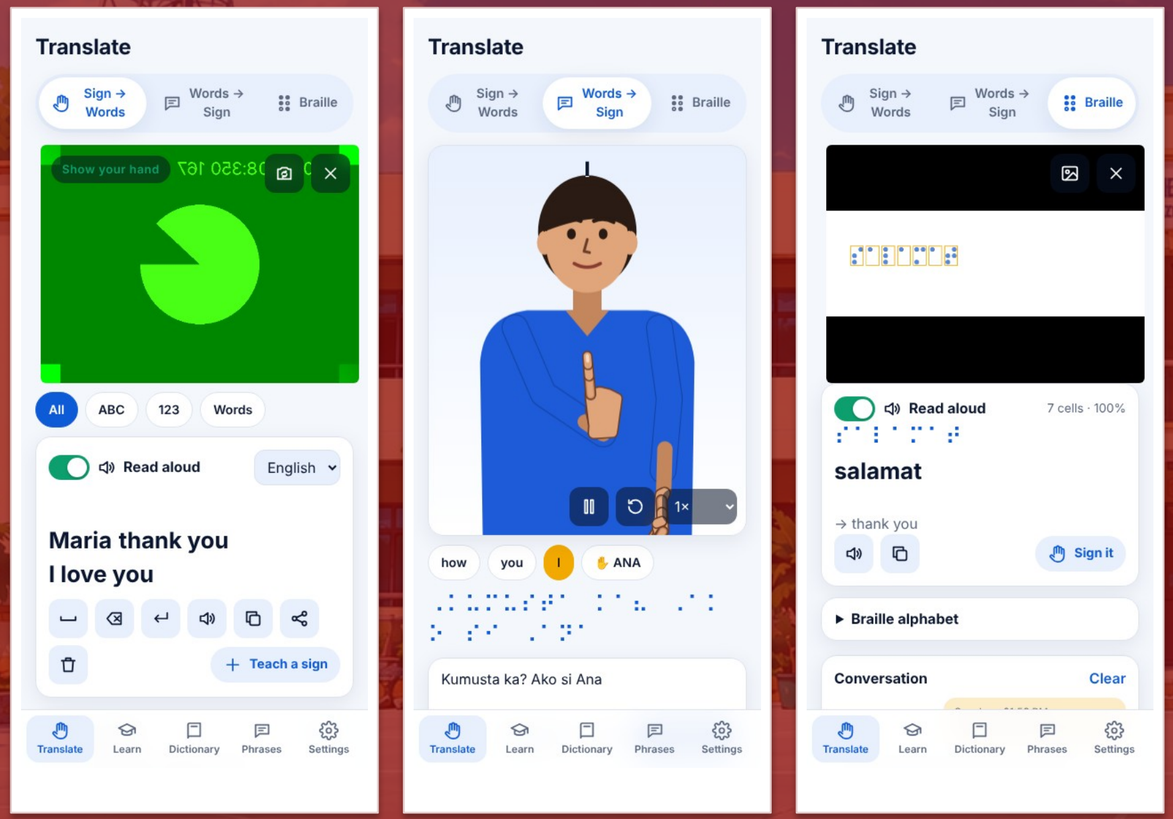


Figure 4. Live translation with Read aloud, the avatar, the braille reader

Hypothesis	Result	Decision
H_o1 accuracy ≤ 90%	99.0%	Rejected
H_o2 detection ≤ 90%	100.0%	Rejected
H_o3 words ≤ 80%	89.2%	Rejected
H_o4 CER ≥ 20%	12.6%	Rejected
H_o5 time > 100 ms	71 + 0.09 ms	Rejected
H_o6 layout faults	0 of 11 sizes	Rejected

All contrast pairs ≥ 4.5 : 1 (WCAG AA). All 32 phrases signed in 3 languages with no fingerspelling.

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

- Geometry and kinematics classify static letters at 99.0% without a dataset.
- Moving and two-handed signs: 100.0%, no false alarms.
- Words 89.2%; taught signs 100.0%; braille 81.7% exact.
- Fast, offline, fits every screen; Read aloud on or off.

Recommendations

- Field-validate with Deaf and blind learners in Albay (ethics review, TAM).
- Calibrate templates with real landmark recordings.
- Review words and signs with Bikol speakers and Deaf FSL users.
- Add facial grammar, Grade 2 braille and a Bikol voice.



QUESTIONS & ANSWERS



THANK YOU